

Formula Sheet Quantum Mechanics 2

Travelling wave solution of Schrödinger equation (travelling in +ve x):

$$\Psi(x, t) = Ae^{i(kx - \omega t)}$$

Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

Time independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = E\Psi$$

Expectation value:

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{A} \Psi dx$$

Eigenvalue equation:

$$\hat{A}\Psi = a\Psi$$

Momentum and kinetic energy operators:

$$-i\hbar \frac{\partial}{\partial x}, \quad -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$$

Transmission and reflection amplitudes:

$$C = \frac{2k_1}{k_1 + k_2}, \quad B = \frac{k_1 - k_2}{k_1 + k_2}$$

Transcendental equations for finite square-well potential:

$$\tan\left(\frac{ka}{2}\right) = \frac{K}{k}, \quad \cot\left(\frac{ka}{2}\right) = -\frac{K}{k}$$

Unnormalised solutions for the harmonic oscillator:

$$\psi_0(u) = e^{-u^2/2}, \quad \psi_1(u) = ue^{-u^2/2}, \\ \psi_2(u) = (2u^2 - 1)e^{-u^2/2}$$

Superposition wave-function:

$$\Phi = \sum_{i=1}^n c_i \Psi_i, \quad c_i = \int_{-\infty}^{\infty} \Psi_i^* \Phi dx$$

Hermitian operator relation:

$$\int_{-\infty}^{\infty} \Psi_b^* \hat{A} \Psi_a dx = \int_{-\infty}^{\infty} \Psi_a (\hat{A} \Psi_b)^* dx$$

Commutator:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

Commutator relationship:

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

Uncertainty principle:

$$\Delta x \Delta p \geq \hbar/2$$

Angular momentum operator:

$$\hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right), \quad [\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

L_z operator:

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

L^2 operator:

$$\hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

Unnormalised Associated Legendre Polynomials:

$$P_0^0(\cos \theta) = 1, \quad P_1^0(\cos \theta) = \cos \theta, \\ P_1^1(\cos \theta) = -\sin \theta, \quad P_1^{-1}(\cos \theta) = (\sin \theta)/2$$

Spherical harmonics:

$$Y_l^m(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$$

Barrier transmission coefficient:

$$T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2Ka}$$